Lab 03 — Build a Loan Amortisation Calculator

Python Programming for Finance · TGS-2026065049 · LO2

Scenario

A relationship manager needs a transparent monthly-instalment calculation including the zero-interest staff-loan case.

Goal

A loan calculator with monthly payment, total repayment and total interest.

Tools

Python, math, csv

Step-by-step

1. Open scenario.md and note the principal, annual rate and term.

2. Run the starter and observe the incomplete result.

```
python starter.py
```

3. Implement the amortisation formula in monthly_payment().

4. Handle a zero annual rate without division by zero.

5. Generate the first 12 schedule rows.

```
python solution.py
```

6. Run the unit tests and save the output.

```
python -m unittest test_solution.py
```

Acceptance criteria

For $500,000 at 2.6% over 25 years, payment is about $2,268.35; a 0% loan returns principal divided by months.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.